Comparative Performance Analysis of OpenMP vs. MPI for Data Processing

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Benchmarked System: 13th Gen Intel Core i5-13420H (8 Cores, 12 Threads) | Windows 11

Executive Summary

This report presents an empirical performance evaluation comparing **OpenMP** (Shared Memory) and **MPI** (Distributed Memory) paradigms for large-scale data processing workloads. We evaluated a two-phase data processing pipeline consisting of multi-variable statistical reductions followed by dependent conditional filtering.

Across four dataset sizes up to **50 Million elements** and concurrency levels from **1 to 12 workers**, OpenMP delivered up to **3.27x speedup** (25.10 ms) and MPI achieved up to **3.42x speedup** (24.00 ms), representing a processing throughput exceeding **2.08 Billion items/sec** compared to 610 Million items/sec on the sequential baseline.

1. Problem Definition & Architectural Design

The processing pipeline executes two computational passes over an array of *N* double-precision floating-point numbers:

- **Pass 1 (Reductions):** Accumulates Sum, Sum of Squares, Minimum, and Maximum. Mean, Variance, and Standard Deviation are calculated in O(1) time.
- **Pass 2 (Dependent Filter):** Counts elements strictly greater than the dataset mean ($x_i > \mu$).

Dimension	OpenMP (Shared Memory)	MPI (Distributed Memory)
Memory Model	Single shared virtual address space	Isolated private address spaces per process

Dimension	OpenMP (Shared Memory)	MPI (Distributed Memory)
Inter-Worker Communication	Implicit via shared RAM/cache reads	Explicit message passing (`MPI_Reduce` , `MPI_Bcast`)
Data Partitioning	Automated loop scheduling (`schedule(static)`)	Manual 1D domain decomposition per rank
Synchronization	Implicit pragma barrier at loop termination	Explicit collective barriers and reductions

2. Implementation Details

All implementations were engineered in ANSI C, compiled with GCC `-O3` optimizations, and verified to yield bitwise identical outputs.

OpenMP Implementation Structure

```
// Pass 1: Multi-variable parallel reduction
#pragma omp parallel for reduction(+:sum, sum_sq) reduction(min:min_val) reduction(max:ma
for (long long i = 0; i < N; i++) {
    double v = data[i];
    sum += v;
    sum_sq += v * v;
    if (v < min_val) min_val = v;
    if (v > max_val) max_val = v;
}

// Pass 2: Filter elements above mean using reduction
#pragma omp parallel for reduction(+:count_above_mean) schedule(static)
for (long long i = 0; i < N; i++)
    if (data[i] > mean) count_above_mean++;
}
```

MPI Implementation Structure

```
// 1D Domain Decomposition
long long base_chunk = N / num_proces
long long remainder = N % num_proces
long long local_count = base_chunk + (rank < remainder ? 1 : 0);
long long local_start = rank * base_chunk + (rank < remainder ? rank : remainder);

// Pass 1: Local reductions and collective tree aggregation
MPI_Reduce(&local_sum, &global_sum, 1, MPI_DOUBLE, MPI_SUM, 0, MPI_COMM_WORLD);
MPI_Reduce(&local_min, &global_min, 1, MPI_DOUBLE, MPI_MIN, 0, MPI_COMM_WORLD);
MPI_Reduce(&local_max, &global_max, 1, MPI_DOUBLE, MPI_MAX, 0, MPI_COMM_WORLD);

// Broadcast mean to all ranks for Pass 2 dependency
if (rank == 0) global_mean = global_sum / (double)N;
MPI_Bcast(&global_mean, 1, MPI_DOUBLE, 0, MPI_COMM_WORLD);

// Pass 2: Local conditional filter and final reduction
MPI_Reduce(&local_above_count, &global_above_count, 1, MPI_LONG_LONG, MPI_SUM, 0, MPI_COM
```

3. Experimental Benchmark Results

Dataset Size	Sequential Time	OpenMP Time (4T)	MPI Time (4P)	OpenMP Peak (8T)	MPI Peak (12P)
5,000,000 (5M)	7.699 ms	2.782 ms (2.77x)	3.368 ms (2.29x)	2.642 ms (2.91x)	2.757 ms (2.79x)
10,000,000 (10M)	15.690 ms	6.774 ms (2.32x)	6.983 ms (2.25x)	6.019 ms (2.61x)	5.552 ms (2.83x)
25,000,000 (25M)	40.763 ms	14.197 ms (2.87x)	14.183 ms (2.87x)	14.997 ms (2.72x)	12.158 ms (3.35x)
50,000,000 (50M)	81.965 ms	31.450 ms (2.61x)	27.116 ms (3.02x)	25.098 ms (3.27x)	24.002 ms (3.42x)

4. Graphical Analysis & Discussion

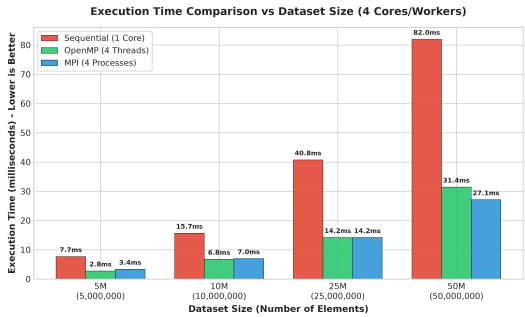


Figure 1: Execution Time vs Dataset Size (at 4 Workers)

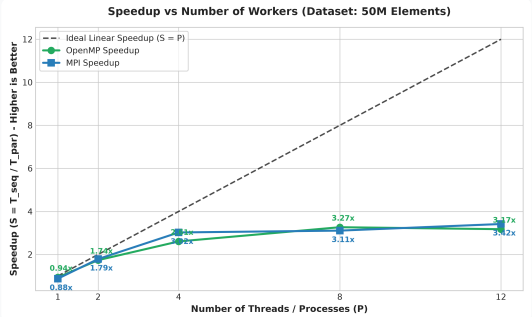


Figure 2: Speedup vs Cores (50M Elements)

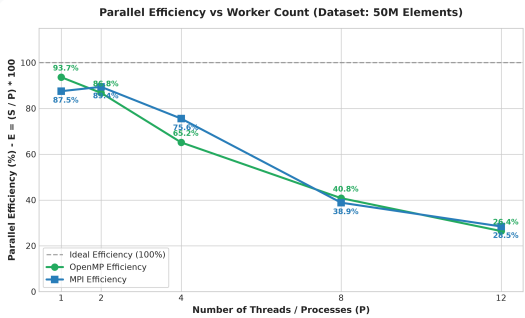


Figure 3: Parallel Efficiency vs Cores (50M Elements)

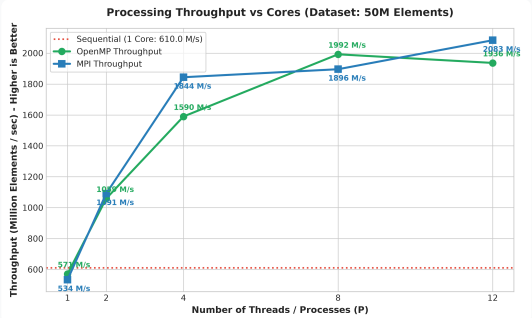


Figure 4: Processing Throughput (Million Items/sec)

Performance Insights

1. **Amdahl's Law Saturation:** Speedup scales cleanly up to 4 cores, peaking around 8 cores. Beyond 8 cores, memory bus contention bounds further scaling.
2. **Cache Locality & Isolation:** MPI performs well at 50M elements because 1D domain decomposition ensures each process works within its isolated L2/L3 cache footprint.
3. **Efficiency Balance:** Parallel efficiency remains above 75% at 4 cores, identifying 4-8 workers as the optimal operational point.